Ex. 2. Answers

2.(a) Linear model $\quad Sales = 569.45 + 140.51\,t$

(5.18) (7.94) t-statistics

$\hat{\alpha}$ $\quad \hat{\beta}$

$\alpha = 0.05$

$t_{n-2;\frac{\alpha}{2}} = t_{10-2;0.025} = 2.306$ $\quad$ Reject H_0, $\alpha \neq 0$ $\quad$ Reject H_0, $\beta \neq 0$

$r = 0.942$ $\quad R^2_{adj} = 1 - \frac{(1-r^2)(n-1)}{n-k-1}$

$R^2_{adj} = 1 - \frac{(1-0.942^2)(10-1)}{10-1-1} = 1 - \frac{(1-0.8874)9}{8}$

$= 1 - 0.1267 = 0.8733 \simeq 0.87$

$F = 63$ $\quad F_{1,9;0.05} = 5.12$ $\quad$ Reject H_0, the model is significant.

$DW = 1.05$ $\quad n=10,\ k'=1,\ d_L = 0.879 < d = 1.05 < d_U = 1.320$

Inconclusive (i.e., no conclusion).

Exponential model $\quad \ln(Sales) = 6.56 + 0.11\,t$

(7.82)

Reject H_0, $\beta \neq 0$

$R^2_{adj} = 0.87$

$F = 61.1$, $\quad F_{1,9;0.05} = 5.12$ $\quad$ Reject H_0, the model is significant.

$DW = 1.12$ $\quad d_L = 0.879 < 1.12 = d < d_U = 1.320$. Inconclusive.

First difference model $\quad Y'_t = X_t - X_{t-1} = Sales_t - Sales_{t-1}$

$= D(sales)$

$D(sales) = 51.54 + 9.21\,t$

(0.42)

$t_{8;0.025} = 2.306$ $\quad$ Fail to reject H_0. Slope is not significant i.e., $\beta = 0$.

$R^2_{adj} = 0.11$

$F = 0.18$, $\quad F_{1,9;0.05} = 5.12$ $\quad$ Fail to reject H_0, the model is not significant.

$DW = 1.56$ $\quad d_U = 1.320 < d = 1.56 < 4 - d_U = 2.680$. No evidence of autocorrelation, i.e., $\rho = 0$.

2(a) (cont'd)

Logistic model $\ln\left(\frac{L}{\hat{Y}_t}-1\right) = \hat{\alpha}+\hat{\beta}t$

$\ln\left(\frac{L}{Sales}-1\right) = 1.14 - 0.34t$ Reject H_0 and conclude $\beta \neq 0$.
(-8.08)

$R^2_{adj} = 0.88$

$F = 65.44$ $F_{1,9;0.05} = 5.12$ Reject H_0 and conclude the model is significant.

$DW = 0.95$ $d_L = 0.879 < d = 0.95 < d_U = 1.320$ No conclusion.

Quadratic model $Y_t = \alpha + \beta_1 t + \beta_2 t^2$

$Sales = 713.29 + 68.59t + 6.54t^2$

$R^2_{adj} = 0.87$

$F = 31.37$ $F_{2,9;0.05} = 4.26$ Reject H_0 and conclude the model is significant.

Cubic model $Y_t = \alpha + \beta_1 t + \beta_2 t^2 + \beta_3 t^3$

$Sales = 1293.06 - 445.64t + 118.03t^2 - 6.76t^3$
(-3.61) (4.63) (-4.42)
Reject H_0 Reject H_0 Reject H_0
Conclude: $\beta_1 \neq 0$ $\beta_2 \neq 0$ $\beta_3 \neq 0$

$R^2_{adj} = 0.96$

$F = 82.95$ $F_{3,9;0.05} = 3.86$ Reject H_0 and conclude that the model is significant.

$DW = 2.42$ $k' = 3$ $d_L = 0.525$ $d_U = 2.016$

$4 - d_U = 1.984 < d = 2.42 < 4 - d_L = 3.475$

No conclusion.

2(a) (cont'd)

Model	R^2_{adj}	RMSE	MAE	MAPE	Theil U
Linear	0.873	143.8	127.6	11.03	0.051
Exponential	0.870	140.2	127.8	9.98	0.05
Logistic	0.877	159.6	118.5	11.43	0.056
Quadratic	0.871	135.7	122.8	9.78	0.048
Cubic	0.965 (1) largest	65.7 (1) smallest	46.6 (1) smallest	4.36 (1) smallest	0.023 (1) closet to 0

Therefore Cubic model is the best model according to all the rankings in various criteria.

Forecasts and prediction interval (based on Cubic model)

$$\hat{X}_t \pm t_{n-2;\frac{\alpha}{2}} S_e \sqrt{1 + \frac{1}{n} + \frac{(t_p - \bar{t})^2}{c_{tt}}}$$

t	Year	lower limit	forecast	upper limit
11	91	1225.7	1679.1	2132.6
12	92	482.5	1265.6	2048.8
13	93	-672.8	601.67	1876.1

↑ t_p values

2(b)

Model	R^2_{adj}	RMSE	MAE	MAPE	Theil U	Rankings
Linear	0.81(2.5)	307.9(4)	235.4(3)	3.97(4)	0.026(4)	17.5
1st Difference	0.07(5)	334.4(5)	237.3(5)	3.96(3)	0.028(5)	23.0
Logistic	0.79(4)	292.5(3)	229.7(1)	3.86(1)	0.025(3)	12.0
Quadratic	0.82(1)	289.1(2)	232.6(2)	3.9(2)	0.024(1.5)	(8.5)
Cubic	0.81(2.5)	285.4(1)	236.5(4)	4.03(5)	0.024(1.5)	14.0

Thus according to the sum of average rankings, Quadratic model has the highest rankings. Therefore Quadratic model is the best amongst all these models.

Forecasts and Prediction Intervals (Quadratic model)

t	Year	lower limit	forecast	upper limit
18	89	5954.15	6838.85	7723.56
19	90	5900.67	6878.75	7856.83
20	91	5813.53	6908.73	8003.93

3.

Year	x_t	SES forecast S_t	$(e_t)^2$	MA3 forecast	$(e_t)^2$
85	1968				
86	1710	1787	$(-77)^2$	1903.33	$(-193.33)^2$
87	1804	1800.6	$(3.4)^2$	1855.00	$(-51)^2$
88	1906	1884.92	$(21.08)^2$	1827.33	$(78.67)^2$
89	1705	1740.98	$(-35.98)^2$	1806.67	$(101.67)^2$
90	1749	1747.40	$(1.60)^2$	1805.00	$(-56)^2$
91	1742	1743.08	$(1.08)^2$	1786.67	$(-44.67)^2$
$\sum$		7683.21 (not 57557)			61634.6556 $\simeq$ 61635

SES $\alpha = 0.8$ $\quad S_t = 0.8\,x_t + 0.2\,S_{t-1}$

$S_{87} = 0.8(1804) + 0.2(1787) = 1800.6$

$S_{88} = 0.8(1906) + 0.2(1800.6) = 1884.92$

$S_{89} = 0.8(1705) + 0.2(1884.92) = 1740.984 \simeq 1740.98$

$S_{90} = 0.8((749) + 0.2(1740.984) = 1747.3968 \simeq 1747.40$

$S_{91} = 0.8((742) + 0.2(1747.3968) = 1743.07936 \simeq 1743.08$

MA3

$S_{86} = \frac{1}{3}(1855 + 1887 + 1968) = 1903.33$

$S_{87} = \frac{1}{3}(1887 + 1968 + 1710) = 1855.0$

$S_{88} = \frac{1}{3}(1968 + 1710 + 1804) = 1827.33$

$S_{89} = \frac{1}{3}(1710 + 1804 + 1906) = 1806.67$

$S_{90} = \frac{1}{3}(1804 + 1906 + 1705) = 1805.0$

$S_{91} = \frac{1}{3}(1906 + 1705 + 1749) = 1786.67$

4. Recall (1.4.1) of Chapter 1.

$$S_T = \alpha X_T + (1-\alpha) S_{T-1} \quad (1)$$
$$S_{T-1} = \alpha X_{T-1} + (1-\alpha) S_{T-2} \quad (2)$$
$$S_{T-2} = \alpha X_{T-2} + (1-\alpha) S_{T-3} \quad (3)$$
$$\vdots$$
$$S_1 = \alpha X_1 + (1-\alpha) S_0 \quad (T)$$

(*) Substitute all (2) to (T) in to (1)

$|\alpha| < 1$

Then $S_T = \alpha \sum_{k=0}^{T-1} (1-\alpha)^k X_{T-k} + (1-\alpha)^T S_0$

As $T \to \infty$, then the last term $\lim_{T\to\infty} (1-\alpha)^T S_0 = 0$

$$E(S_T) = E\left[\alpha \sum_{k=0}^{\infty} (1-\alpha)^k X_{T-k}\right] = \alpha \sum_{k=0}^{\infty} (1-\alpha)^k E(X_{T-k})$$

$$= \alpha \cdot \frac{1}{1-(1-\alpha)} E(X_{T-k}) = E(X_{T-k})$$

Similarly, $S_T^{[2]} = \alpha S_T + (1-\alpha) S_{T-1}^{[2]}$

Use (*) steps, we can show $E(S_T^{[2]}) = E(S_T) - \frac{\beta}{\alpha} b_2$

For $X_t = b_1 + b_2 t + \varepsilon_t$, then $E(X_t) = b_1 + b_2 t$

and $X_{t-k} = b_1 + b_2(t-k) + \varepsilon_t$, then $E(X_{t-k}) = b_1 + b_2(t-k)$

$$E(S_T^{[2]}) = \alpha \sum_{k=0}^{\infty} (1-\alpha)^k \left[E(X_{T-k}) - \frac{1-\alpha}{\alpha} b_2\right]$$

$$= \alpha \sum_{k=0}^{\infty} (1-\alpha)^k \left[b_1 + b_2(T-k) - \frac{1-\alpha}{\alpha} b_2\right]$$

$$= \underbrace{b_1 + b_2 T - \frac{1-\alpha}{\alpha} b_2} - \frac{1-\alpha}{\alpha} b_2$$

$$= E(S_T) - \frac{1-\alpha}{\alpha} b_2.$$